





Testing Spring Boot Applications Demystified

Best Practices, Common Pitfalls, and Real-World Strategies

Java User Group St. Gallen 22.10.2025

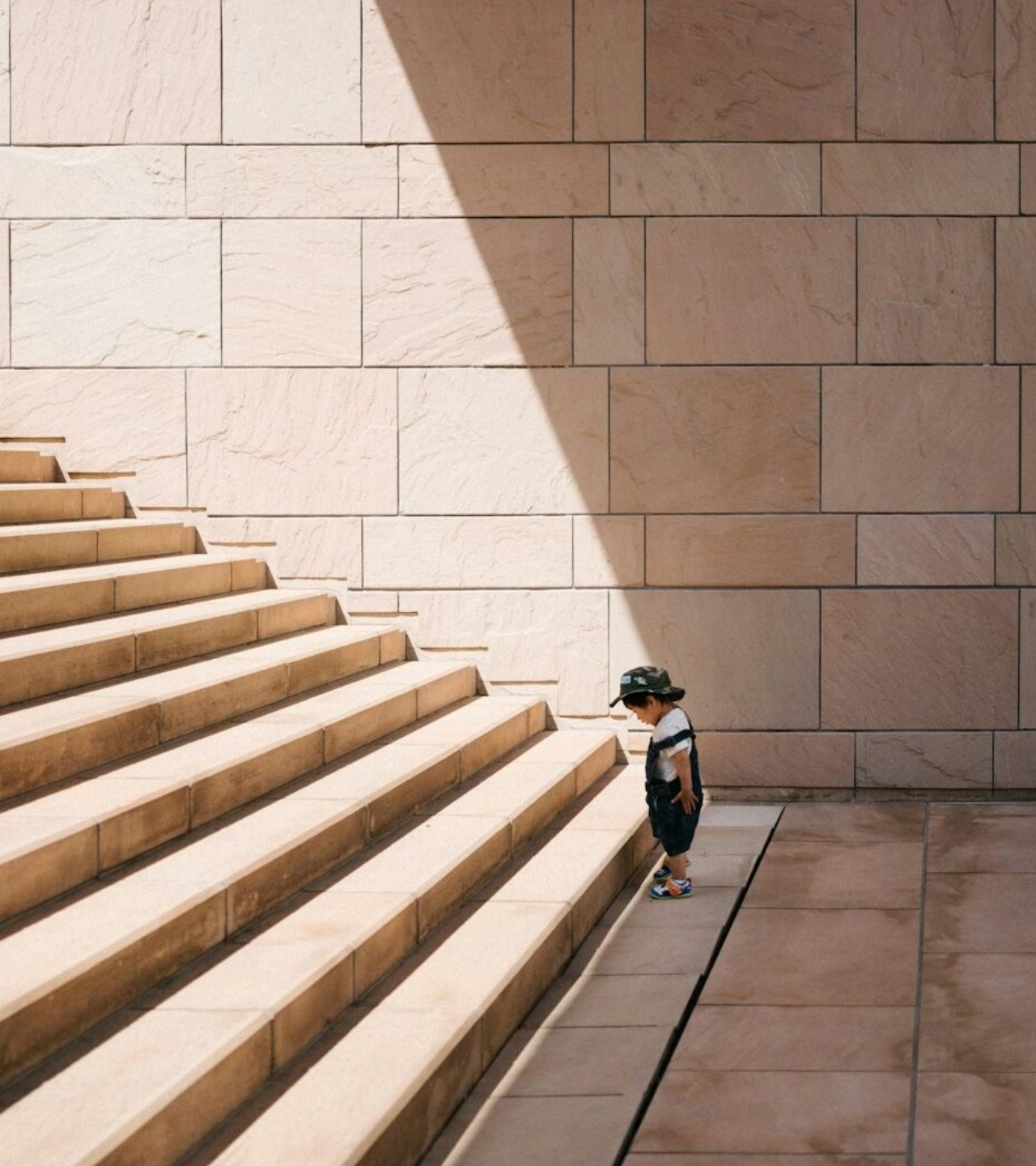
Philip Riecks - [PragmaTech GmbH](#) - [@rieckpil](#)

Participate During the Talk & Win Prizes

Go to menti.com on your phone or laptop and use the code **2298 6846** to submit answers for the quizzes and add your questions during the talk.

If you want to take part in the raffle, please add your first name and the first letter of your last name to each **text-filed submission** like "What's your preferred % for code coverage (Philip R.)" - not for the single choice questions.





Getting Started with Testing

How It Started

Getting Used To Testing At Work

 **theRealLeadDev** requested changes 1 minute ago

[View changes](#)

theRealLeadDev left a comment

Changes look good!

But where are the tests? I can see no changes/new files within `src/test/java` .

We can't integrate it that way. Please add some tests and ping me again.



Goals For This Talk

- Lay the foundation for your Spring Boot testing success
- Introduction to Spring Boot's excellent test support
- Showcase a mix of best practices and early pitfalls
- Convince you that testing is not an afterthought



About Philip

- Self-employed developer from Herzogenaurach, Germany (Bavaria) 🍺
- Blogging & content creation with a focus on testing Java and specifically Spring Boot applications 🌿
- Founder of PragmaTech GmbH - Enabling Developers to Frequently Deliver Software with More Confidence 🚀
- Enjoys writing tests 📱
- @rieckpil on various platforms



Agenda

- Introduction
- Why Test Software?
- Testing with Spring Boot
 - Spring Boot Testing 101 & Unit Testing
 - Sliced Testing
 - Integration & E2E Testing
- Spring Boot Testing Best Practices
- Common Spring Boot Testing Pitfalls
- Summary & Outlook

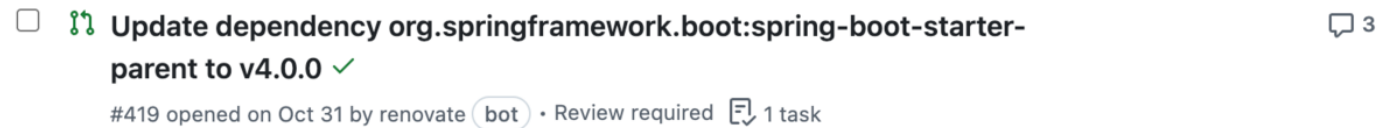


Why Test Software?



My Overall Northstar

Imagine seeing this pull request on a Friday afternoon:



How confident are you to merge this major Spring Boot upgrade and deploy it to production once the pipeline turns green?

Good tests don't just catch bugs - they give you the confidence to say "yes" without hesitation.



Why Test Software? (continued)

- **Shift Left** - Catch issues earlier than the customers
- **Confidence in Code Changes** - Help new team members to onboard faster
- **Catch Bugs Early** - Reduce the (\$) cost of bugs in production
- **Documentation** - Single point of truth for implemented business logic
- **Regression Prevention** - Prevent existing functionality from breaking
- **Become more Productive** - Enable faster development cycles
- **Use it as a Playground** - Explore new technologies via tests



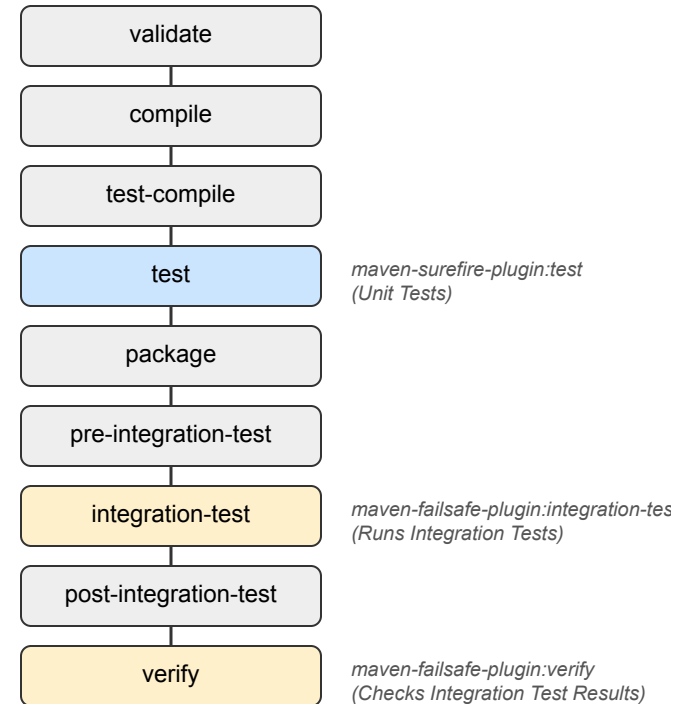
Spring Boot Testing 101 & Unit Testing



Maven Build Lifecycle

- **Maven Surefire Plugin** for unit tests: default postfix `*Test` (e.g. `CustomerTest`)
- **Maven Failsafe Plugin** for integration tests: default postfix `*IT` (e.g. `CheckoutIT`)
- Reason for splitting: different **parallelization** options, better **organisation**

Running `./mvnw verify`



Legend:

- Surefire Execution Phase
- Failsafe Execution Phases



Gradle Build Lifecycle

- Unit tests are run during the `test` task
- To separate integration tests, we need a custom Gradle task, as this structure is **not part** of default Gradle lifecycle
- We **need to configure** the `integrationTest` task manually in our `build.gradle`:

```
1 // Sample configuration from the Gradle docs
2 tasks.register('integrationTest', Test) {
3     description = 'Runs integration tests.'
4     group = 'verification'
5
6     // ...
7     shouldRunAfter test
8
9     useJUnitPlatform()
10 }
```



Spring Boot Starter Test

- aka. "Testing Swiss Army Knife"

```
1 <dependency>
2   <groupId>org.springframework.boot</groupId>
3   <artifactId>spring-boot-starter-test</artifactId>
4   <scope>test</scope>
5 </dependency>
```

- Batteries-included for testing by transitively including popular testing libraries
 - JUnit
 - Mockito
 - Assertion libraries: AssertJ, Hamcrest, XMLUnit, JSONAssert, Awaitility




```
1 ./mvnw dependency:tree
2 [INFO] ...
3 [INFO] +- org.springframework.boot:spring-boot-starter-test:jar:3.5.6:test
4 [INFO] | +- org.springframework.boot:spring-boot-test:jar:3.5.6:test
5 [INFO] | +- org.springframework.boot:spring-boot-test-autoconfigure:jar:3.5.6:test
6 [INFO] | +- com.jayway.jsonpath:json-path:jar:2.9.0:test
7 [INFO] | +- jakarta.xml.bind:jakarta.xml.bind-api:jar:4.0.2:test
8 [INFO] | | \- jakarta.activation:jakarta.activation-api:jar:2.1.4:test
9 [INFO] | +- net.minidev:json-smart:jar:2.5.2:test
10 [INFO] | | \- net.minidev:accessors-smart:jar:2.5.2:test
11 [INFO] | | \- org.ow2.asm:asm:jar:9.7.1:test
12 [INFO] | +- org.assertj:assertj-core:jar:3.27.4:test
13 [INFO] | | \- net.bytebuddy:byte-buddy:jar:1.17.7:test
14 [INFO] | +- org.awaitility:awaitility:jar:4.3.0:test
15 [INFO] | +- org.hamcrest:hamcrest:jar:3.0:test
16 [INFO] | +- org.junit.jupiter:junit-jupiter:jar:5.12.2:test
17 [INFO] | | +- org.junit.jupiter:junit-jupiter-api:jar:5.12.2:test
18 [INFO] | | | +- org.junit.platform:junit-platform-commons:jar:1.12.2:test
19 [INFO] | | | \- org.apiguardian:apiguardian-api:jar:1.1.2:test
20 [INFO] | | +- org.junit.jupiter:junit-jupiter-params:jar:5.12.2:test
21 [INFO] | | \- org.junit.jupiter:junit-jupiter-engine:jar:5.12.2:test
22 [INFO] | | \- org.junit.platform:junit-platform-engine:jar:1.12.2:test
23 [INFO] | +- org.mockito:mockito-core:jar:5.16.0:test
24 [INFO] | | +- net.bytebuddy:byte-buddy-agent:jar:1.17.7:test
25 [INFO] | | \- org.objenesis:objenesis:jar:3.3:test
26 [INFO] | +- org.mockito:mockito-junit-jupiter:jar:5.16.0:test
27 [INFO] | +- org.skyscreamer:jsonassert:jar:1.5.3:test
28 [INFO] | | \- com.vaadin.external.google:android-json:jar:0.0.20131108.vaadin1:test
29 [INFO] | +- org.springframework:spring-core:jar:6.2.11:compile
30 [INFO] | | \- org.springframework:spring-jcl:jar:6.2.11:compile
31 [INFO] | +- org.springframework:spring-test:jar:6.2.11:test
32 [INFO] | \- org.xmlunit:xmlunit-core:jar:2.10.4:test
```



What's Inside the Testing Swiss Army Knife?

- **JUnit** (currently 5, later 6): Java's de-facto standard testing framework and foundation.
- **Mockito**: Creating mock objects to simulate dependencies and verify interactions.
- **AssertJ**: Provides fluent, chainable, and readable assertions.
- **Hamcrest**: Offers flexible matchers for creating custom assertions.
- **JSONAssert**: Compares JSON strings with flexible matching options.
- **JsonPath**: Extracts and queries data from JSON similar to XPath.
- **XMLUnit**: Compares and validates XML documents.
- **Awaitility**: Handles asynchronous testing with fluent conditions.



Unit Testing Spring Boot Applications 101

- **Core Concept:** Test individual components (classes, methods) in complete isolation from their dependencies.
- **Confidence Gained:** Provides logarithmic verifications, ensuring that the smallest parts of your code work as expected under various conditions.
- **Best Practices:** Focus on a single unit of work.
- **Pitfalls:** Requires a well-thought-out class design. Poor design can lead to testing overly complex "god classes," making tests difficult to write and maintain.
- **Tools:** JUnit (or Spock, TestNG, etc.), Mockito and assertion libraries like AssertJ or Hamcrest.



Unit Testing with Spring Boot

- Provide collaborators from outside (dependency injection) -> no `new` inside our code
- Develop small, single responsibility classes
- Test only the public API of our class
- Verify behavior not implementation details
- TDD can help design (better) classes



Unify Test Structure

- Use a consistent test method naming: givenWhenThen, shouldWhen, etc.
- Structure test for the Arrange/Act/Assert test setup

```
1 @Test
2 void should_When_() {
3
4     // Arrange
5     // ... setting up objects, data, collaborators, etc.
6
7     // Act
8     // ... performing the action to be tested on the class under test
9
10    // Assert
11    // ... verifying the expected outcome
12 }
```



```
1 @ExtendWith(MockitoExtension.class)
2 class CustomerServiceTest {
3
4     @Mock
5     private CustomerRepository customerRepository;
6
7     @InjectMocks
8     private CustomerService customerService;
9
10    @Test
11    void shouldCreateNewCustomerWhenNameDoesNotExist() {
12
13        when(customerRepository.findByCustomerName("duke"))
14            .thenReturn(Optional.empty());
15
16        when(customerRepository.save(any(CustomerEntity.class)))
17            .thenAnswer(invocation -> {
18                CustomerEntity storedCustomer = invocation.getArgument(0);
19                storedCustomer.setId("42");
20                return storedCustomer;
21            });
22
23        String customerId = customerService.createNewCustomer("duke");
24
25        assertThat(customerId).isEqualTo("42");
26    }
27 }
```



Unit Testing Has Limits

Consider this sample REST controller, what could we verify with a unit test?

```
1 @RestController
2 @RequestMapping("/api/customers")
3 public class CustomerController {
4
5     private final CustomerService customerService;
6
7     public CustomerController(CustomerService customerService) {
8         this.customerService = customerService;
9     }
10
11     @PostMapping
12     public ResponseEntity<Void> createNewCustomer(@Validated CustomerCreationRequest payload, UriComponentsBuilder uriBuilder) {
13
14         String customerId = customerService.createNewCustomer(payload.firstName());
15
16         UriComponents uriComponents = uriBuilder
17             .path("/api/customers/{id}")
18             .buildAndExpand(customerId);
19
20         return ResponseEntity.created(uriComponents.toUri()).build();
21     }
22 }
```



```
1 @ExtendWith(MockitoExtension.class)
2 class CustomerControllerUnitTests {
3
4     @Mock
5     private CustomerService customerService;
6
7     @InjectMocks
8     private CustomerController customerController;
9
10    @Test
11    void shouldCreateCustomerWhenPayloadRequestIsValid() {
12        when(customerService.createNewCustomer(anyString()))
13            .thenReturn("42");
14
15        ResponseEntity<Void> result = customerController.createNewCustomer(
16            new CustomerCreationRequest("Java", "Duke", "duke@jug.ch"),
17            UriComponentsBuilder.newInstance()
18        );
19
20        assertThat(result.getStatusCode().value())
21            .isEqualTo(201);
22        assertThat(result.getHeaders().getLocation().toString())
23            .isEqualTo("/api/customers/42");
24    }
25 }
```

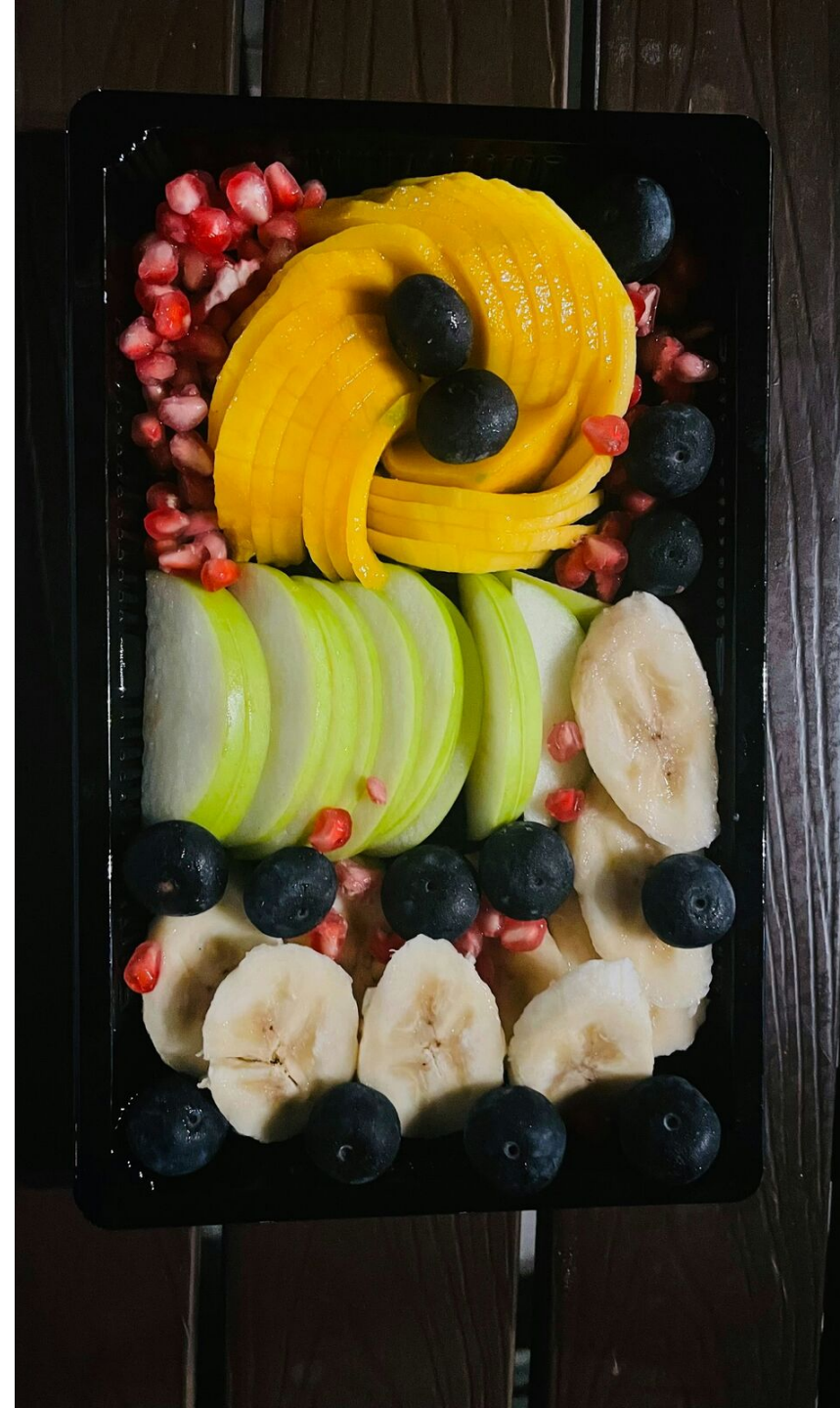


Things We Can't Cover with a Unit Test

- **Request Mapping:** Does HTTP GET `/api/customers/{id}` actually resolve to our desired method?
- **Validation:** Will incomplete request bodys result in a 400 bad request or return an accidental 200?
- **Serialization:** Are we JSON objects serialized and deserialized correctly?
- **Headers:** Are we setting `Content-Type` or custom headers correctly?
- **Security:** Are we Spring Security configuration and other authorization checks enforced?

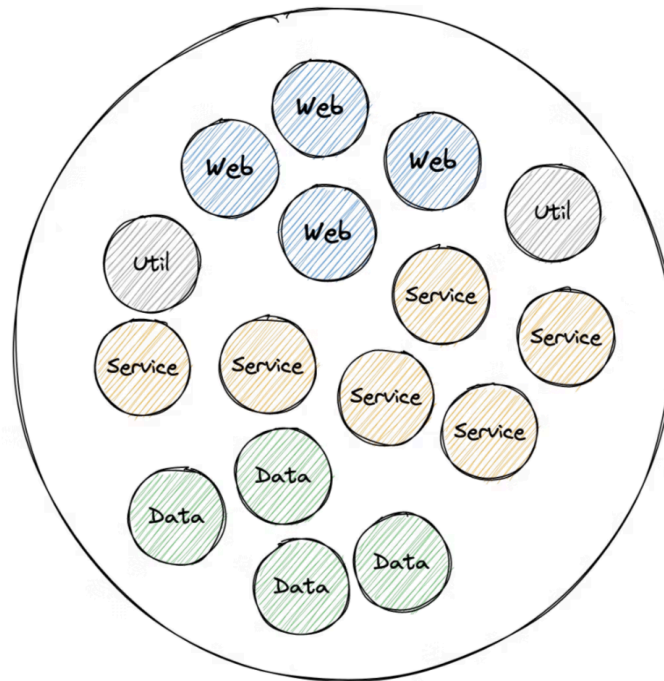


Sliced Testing



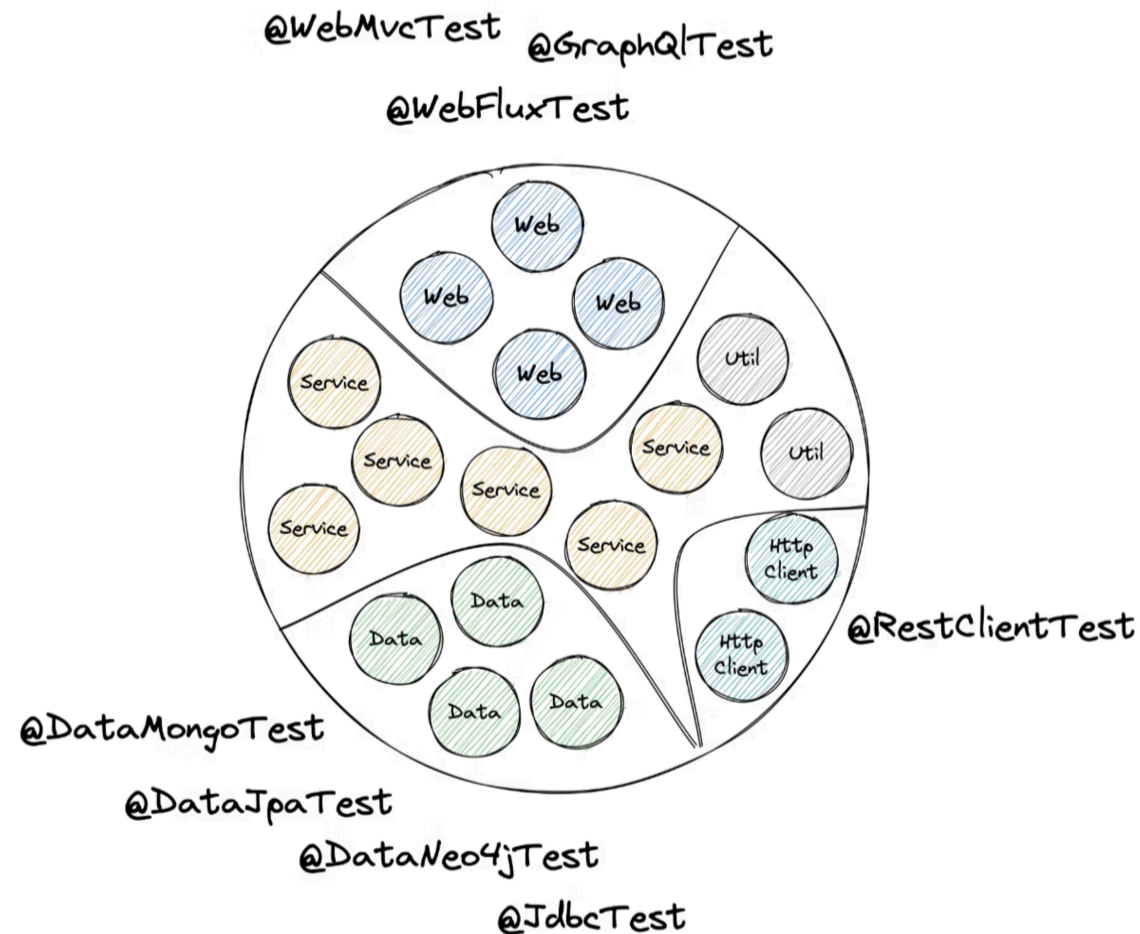
A Typical Spring **ApplicationContext**

Our application context consists of many different components (Spring beans):



We Can Slice It!

Spring Boot allows to load only specific parts (slices) of the application context:



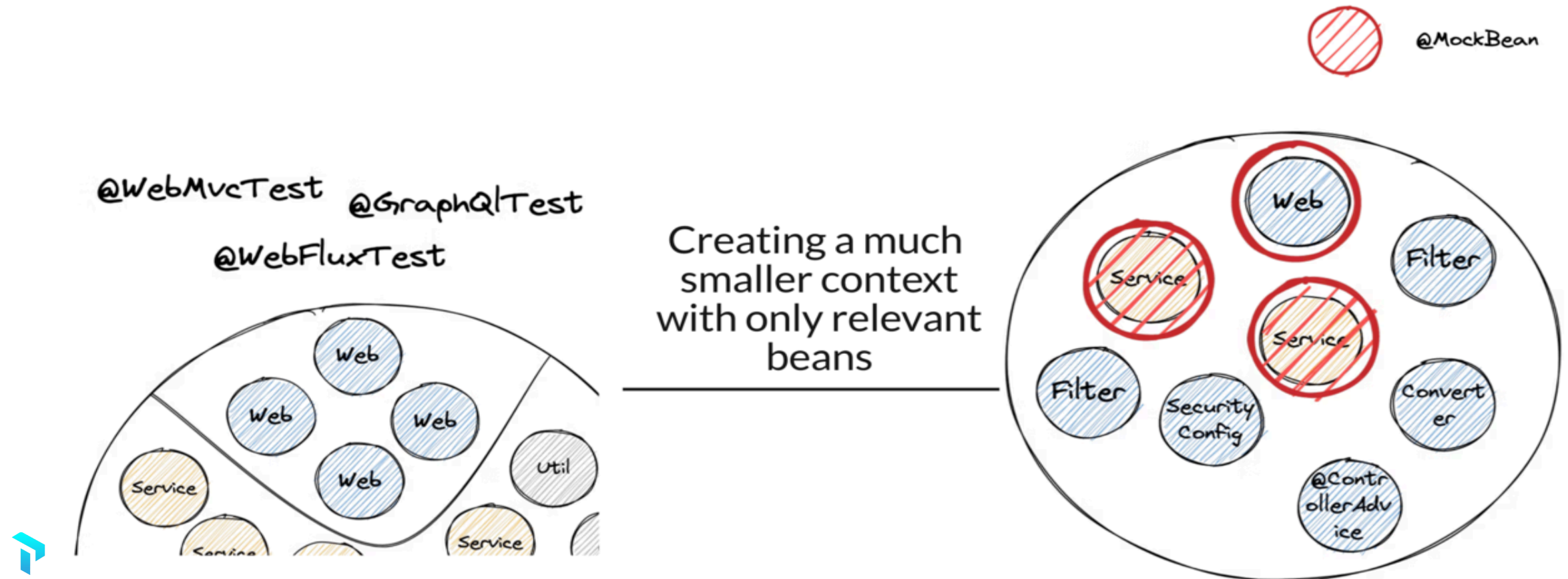
Sliced Testing Spring Boot Applications 101

- **Core Concept:** Test a specific "slice" or layer of your application by loading a minimal, relevant part of the Spring `ApplicationContext`.
- **Confidence Gained:** Helps validate parts of your application where pure unit testing is insufficient, like the web, messaging, or data layer.
- **Prominent Examples:** Web layer (`@WebMvcTest`) and database layer (`@DataJpaTest`)
- **Pitfalls:** Requires careful configuration to ensure only the necessary slice of the context is loaded.
- **Tools:** JUnit, Mockito, Spring Test, Spring Boot, Testcontainers



Slicing in Action

Spring Boot's test slice component scanning will only include relevant beans in the sliced context. We need to provide or mock beans that are not part of the slice:



Slicing Example: @WebMvcTest

- Testing the web layer in isolation and only load the beans we need
- `MockMvc` : Mocked servlet environment with HTTP semantics
- See `WebMvcTypeExcludeFilter` for included Spring beans

```
1 @WebMvcTest(CustomerController.class)
2 class CustomerControllerTest {
3
4     @Autowired
5     private MockMvc mockMvc;
6
7     @MockitoBean
8     private CustomerService customerService;
9
10 }
```



Common Test Slices

- `@WebMvcTest` / `@WebFluxTest` - Controller layer
- `@DataJpaTest` / `@JdbcTest` - Persistence layer
- `@JsonTest` - JSON serialization/deserialization
- `@RestClientTest` - RestTemplate testing
- etc.



@DataCouchbaseTest
@DataLdapTest
@RestClientTest
@JsonTest
@JooqTest
@WebFluxTest
@DataRedisTest
@WebMvcTest
@DataMongoTest
@DataCassandraTest
@GraphQLTest
@DataNeo4jTest
@WriteYourOwn*
@DataJpaTest
@SqsTest
@JdbcTest
@DataElasticsearchTest



Integration Testing

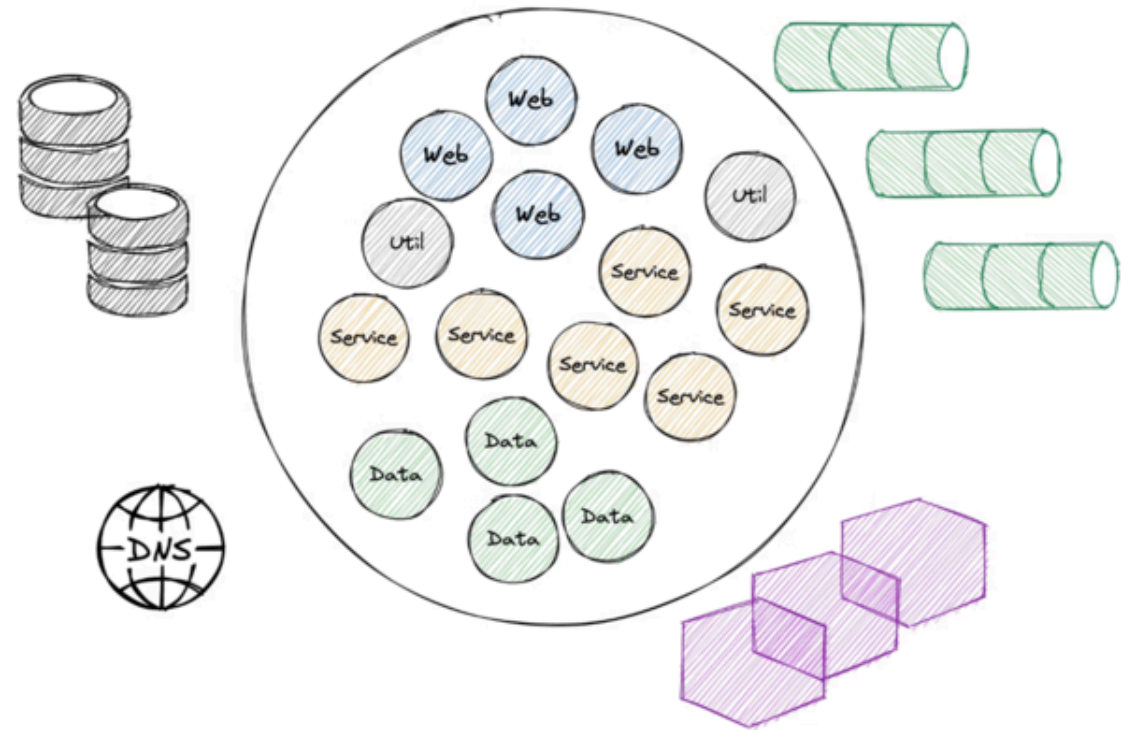



```
@SpringBootTest
class ApplicationTest {

    @Test
    void contextLoads() {

    }

}
```



Integration Testing Spring Boot Applications 101

- **Core Concept:** Start the entire Spring application context, often on a random local port, and test the application through its external interfaces (e.g., REST API).
- **Confidence Gained:** Validates the integration of all internal components working together as a complete application.
- **Best Practices:** Use `@SpringBootTest` to run the app on a local port.
- **Pitfalls:** Slower to run than unit or sliced tests. Managing the lifecycle of dependent services can be complex.
- **Tools:** JUnit, Mockito, Spring Test, Spring Boot, Testcontainers, WireMock (for mocking external HTTP services), Selenium (for browser-based UI testing)



Starting the Entire Context

- Provide external infrastructure with [Testcontainers](#)
- Start Tomcat with: `@SpringBootTest(webEnvironment = WebEnvironment.RANDOM_PORT)`
- Consider WireMock/MockServer for stubbing external HTTP services
- Test controller endpoints via: `MockMvc`, `WebTestClient`, `TestRestTemplate`
- Speed up builds with Spring Test `TestContext` caching



Provide External Infrastructure with Testcontainers

Running infrastructure components (databases, message brokers, etc.) in Docker containers for our tests becomes a breeze with [Testcontainers](#):

```
1 @Container
2 @ServiceConnection
3 static PostgreSQLContainer<?> postgres = new PostgreSQLContainer<>("postgres:16-alpine")
4     .withDatabaseName("testdb")
5     .withUsername("test")
6     .withPassword("test")
7     .withInitScript("init-postgres.sql");
```

This gives us an ephemeral PostgreSQL database for our tests:

```
1 $ docker ps
2 CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS                               NAMES
3 a958ee2887c6   postgres:16-alpine                "docker-entrypoint.s..." 10 seconds ago Up 9 seconds  0.0.0.0:32776->5432/tcp, [::]:32776->5432/tcp affectionate_cannon
4 ad0f804068dc   testcontainers/ryuk:0.12.0         "/bin/ryuk"              10 seconds ago Up 9 seconds  0.0.0.0:32775->8080/tcp, [::]:32775->8080/tcp testcontainers-ryuk-1f9f76a6-46d4-4e19-85c1-e8364da12804
```



Stub External HTTP Services with WireMock

Consider [WireMock](#) to stub external HTTP services during tests.

- Run as in-memory service or Docker container to simulate connected HTTP services
- Simulate failures, slow responses, etc.
- Stateful setups possible (scenarios): first request fails, then succeeds
- Override HTTP clients to connect to the WireMock server during tests

```
1 wireMockServer.stubFor(  
2     get(urlPathEqualTo("/api/books/" + isbn))  
3         .willReturn(aResponse()  
4             .withHeader(HttpHeaders.CONTENT_TYPE, MediaType.APPLICATION_JSON_VALUE)  
5             .withBodyFile("book-response-success.json"))  
6 );
```



Starting the Entire Spring Context - Version 1

- We access the application over HTTP like a user, the test and context run in separate threads (no `@Transactional` rollback), requires HTTP authentication

```
1 @SpringBootTest(webEnvironment = SpringBootTest.WebEnvironment.RANDOM_PORT)
2 class ApplicationServletContainerIT {
3
4     @LocalServerPort
5     private int port; // <-- we're running on a real port
6
7     @Test
8     void contextLoads(@Autowired WebTestClient webTestClient) {
9         webTestClient
10             .get()
11             .uri("/api/customers")
12             .header("Authorization", "Basic " + Base64.getEncoder().encodeToString("user:dummy".getBytes()))
13             .exchange()
14             .expectStatus()
15             .isOk();
16     }
17 }
```



Starting the Entire Spring Context - Version 2

- The test and the context run in the same thread, hence we can rollback with `@Transactional` and simply override the security context with `@WithMockUser`

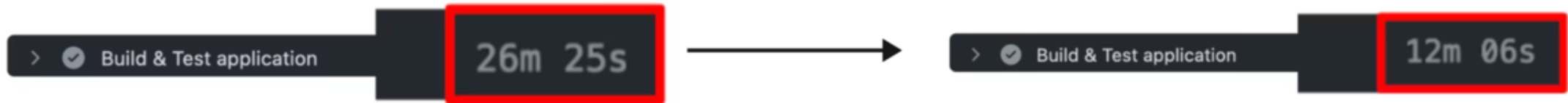
```
1 @SpringBootTest
2 // which is @SpringBootTest(webEnvironment = SpringBootTest.WebEnvironment.MOCK)
3 @AutoConfigureMockMvc
4 class ApplicationMockWebIT {
5
6     // @LocalServerPort
7     // private int port; <-- this would fail the test, there is no local port occupied
8
9     @Test
10    @WithMockUser
11    void givenCustomersThenReturnListForAuthenticatedUser(@Autowired MockMvc mockMvc) throws Exception {
12        mockMvc
13            .perform(get("/api/customers")
14                .header(ACCEPT, APPLICATION_JSON))
15            .andExpect(status().is(200))
16            .andExpect(content().contentType(APPLICATION_JSON))
17            .andExpect(jsonPath("$.size()", is(1)));
18    }
19 }
```



The Need for Speed - Reduce Build Times with Context Caching

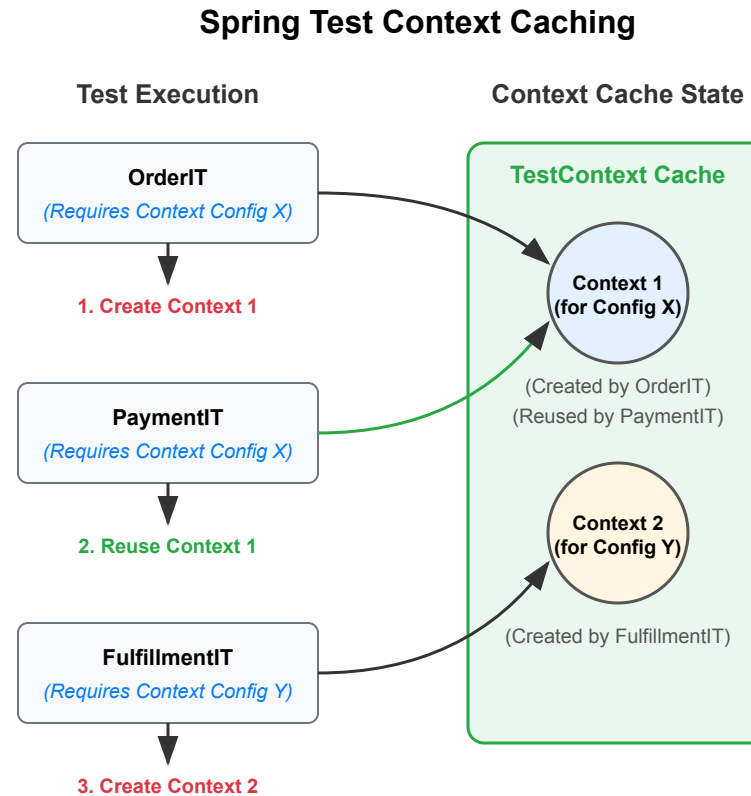
- **The problem:** Integration tests require a started & initialized Spring `ApplicationContext`, which takes time to start
- **The solution:** Spring Test `TestContext` caching, caches an already started Spring `ApplicationContext` for later reuse
- This feature is part of Spring Test (part of every Spring Boot project via `spring-boot-starter-test`)

Speed improvement example:



Caching is King

How the caching mechanism works:



How the Cache Key is Built

```
1 // DefaultContextCache.java
2 private final Map<MergedContextConfiguration, ApplicationContext> contextMap =
3     Collections.synchronizedMap(new LruCache(32, 0.75f));
```

This goes into the cache key (`MergedContextConfiguration`):

- `activeProfiles ( @ActiveProfiles )`
- `contextInitializersClasses ( @ContextConfiguration )`
- `propertySourceLocations ( @TestPropertySource )`
- `propertySourceProperties ( @TestPropertySource )`
- `contextCustomizer ( @MockitoBean , @MockBean , @DynamicPropertySource , ...)`



Identify Context Restarts - Visually

<default package>

OrderControllerReusableTest

shouldAllowAccessForAnonymousUsers()

CustomerControllerReusableTest

shouldAllowAccessForAnonymousUsers()

233 ms

228 ms

228 ms

5 ms

5 ms

Tests passed: 2 of 2 tests - 233 ms

/Library/Java/JavaVirtualMachines/zulu-17.jdk/Contents/Home/bin/java ...

2022-05-09 11:24:33.653 DEBUG --- [main] o.s.test.context.BootstrapUtils :
Instantiating CacheAwareContextLoaderDelegate from class [org.springframework.test.context.cache
.DefaultCacheAwareContextLoaderDelegate]

2022-05-09 11:24:33.662 DEBUG --- [main] o.s.test.context.BootstrapUtils :
Instantiating BootstrapContext using constructor [public org.springframework.test.context
.support.DefaultBootstrapContext(java.lang.Class,org.springframework.test.context
.CacheAwareContextLoaderDelegate)]

2022-05-09 11:24:33.687 DEBUG --- [main] o.s.test.context.BootstrapUtils :
Instantiating TestContextBootstrapper for test class [de.riECKpil.talks.introduction
web

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:: Spring Boot :: (v2.6.7)

2022-05-09 11:25:51.412 INFO 43090 --- [main] d.r.t.introduction.OrderControllerTest
: Starting OrderControllerTest using Java 17 on Philips-MacBook-Pro.local with PID 43090
(started by rieckpil in /Users/riECKpil/Development/git/talks/fixing-a-broken-window)



Identify Context Restarts - with Logs

```
<logger name="org.springframework.test.context" level="TRACE" />
```

```
2022-05-05 14:27:38.136 DEBUG 42500 --- [          main] org.springframework.test.context.cache : Spring test
ApplicationContext cache statistics: [DefaultContextCache@68e62b3b size = 1, maxSize = 32, parentContextCount = 0,
hitCount = 11, missCount = 1]
2022-05-05 14:27:38.136 DEBUG 42500 --- [          main] tractDirtiesContextTestExecutionListener : After test class:
context [DefaultTestContext@325bb9a6 testClass = ApplicationIT, testInstance = [null], testMethod = [null], testException
= [null], mergedContextConfiguration = [WebMergedContextConfiguration@1d12b024 testClass = ApplicationIT, locations =
'{}', classes = '{class de.riECKpil.talks.Application}', contextInitializerClasses = '[]', activeProfiles = '{}',
propertySourceLocations = '{}', propertySourceProperties = '{org.springframework.boot.test.context
.SpringBootTestContextBootstrapper=true, server.port=0}', contextCustomizers = set[org.springframework.boot.test.context
.filter.ExcludeFilterContextCustomizer@5215cd9a, org.springframework.boot.test.json
.DuplicateJsonObjectContextCustomizerFactory$DuplicateJsonObjectContextCustomizer@9257031, org.springframework.boot.test
```



Identify Context Restarts - with Tools



An [open-source Spring Test utility](#) that provides visualization and insights for Spring Test execution, with a focus on Spring context caching statistics.

Overall goal: Identify optimization opportunities in your Spring Test suite to speed up your builds and ship to production faster and with more confidence.



The Final Boss

Developers tend to consult AI/StackOverflow for integration test issues and often copy advice from the internet without knowing the implications:

```
1 @SpringBootTest
2 @DirtiesContext
3 // this instructs Spring to remove the context from the cache
4 // and rebuild a new context on every request
5 public abstract class AbstractIntegrationTest {
6
7 }
```

The setup above will **disable** the context caching feature and slow down the builds significantly!



Spot the Issues for Context Caching

```
@DirtiesContext
@Testcontainers
@ActiveProfiles("integration-test")
@SpringBootTest(webEnvironment = SpringBootTest.WebEnvironment.RANDOM_PORT)
abstract class AbstractIntegrationTest {
```

What's "bad" for
context caching
here?

```
@ActiveProfiles("integration-test")
@Import(SomeTestConfiguration.class)
@ContextConfiguration(initializers = CustomInitializer.class)
@SpringBootTest(properties = {"features.login-enabled=true", "custom.message=duke42"})
class ShowcaseIT {

    @MockBean
    private OrderService orderService;

    @SpyBean
    private CustomerService customerService;

    @Test
    void shouldInitializeContext(@Autowired ApplicationContext applicationContext) {
        assertThat(applicationContext)
            .isNotNull();
    }
}
```



Outlook to Spring Framework 7: Pausing of Test Contexts

See the release notes of [Spring Framework 7.0.0 M7](#).

Pausing of Test Application Contexts

The Spring TestContext framework is caching application context instances within test suites for faster runs. As of Spring Framework 7.0, we now pause test application contexts when they're not used.

This means an application context stored in the context cache will be stopped when it is no longer actively in use and automatically restarted the next time the context is retrieved from the cache.

Specifically, the latter will restart all auto-startup beans in the application context, effectively restoring the lifecycle state.



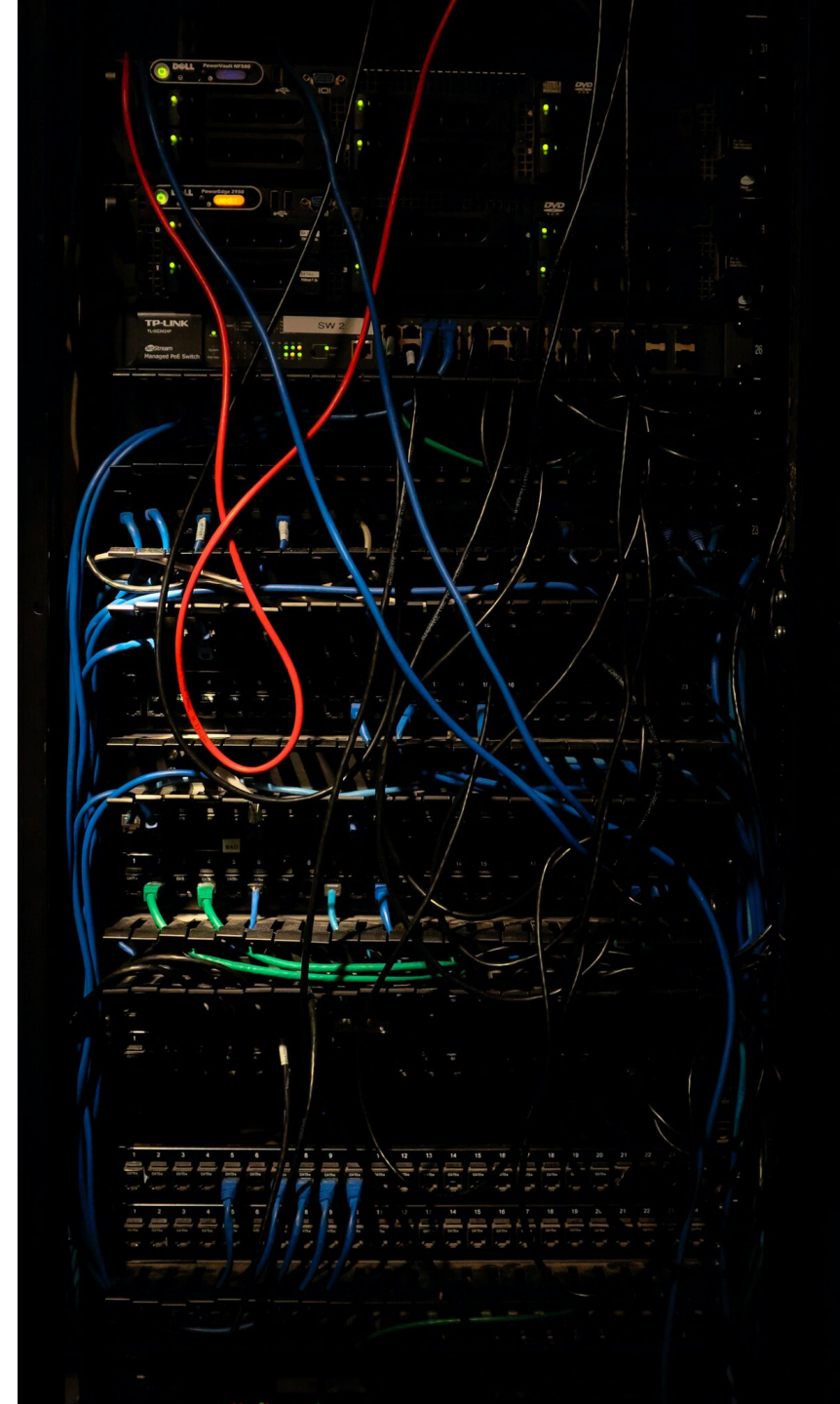
Make the Most of the Caching Feature

- Avoid `@DirtyiesContext` when possible, especially central places
- Understand how the cache key is built
- Monitor and investigate the context restarts
- Align the number of unique context configurations for your test suite

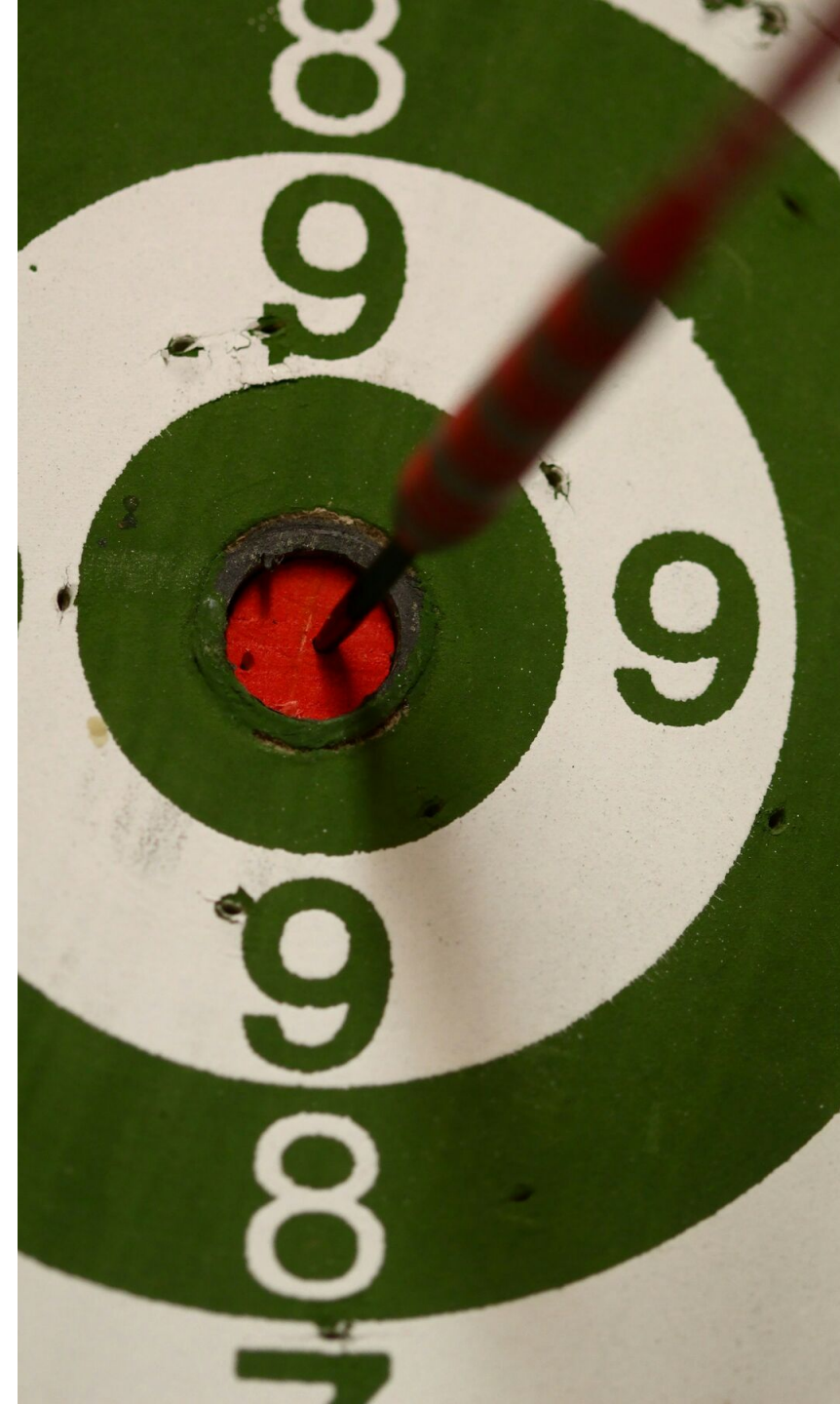


E2E Testing - the Holy Grail of Confidence

- For applications involving a UI consider tools like Selenium, Selenide, Cypress, Playwright, etc.
- Detect issues that only appear in production-like environments, also for downstream systems
- Start with a QA/DEV environment
- Consider Canary Testing and run your E2E tests regularly with a cron-like setup
- Challenges: authentication, test data management, environment stability, flakiness



Spring Boot Testing Best Practices



Best Practice 1: Test Parallelization

Goal: Reduce build time and get faster feedback

Requirements:

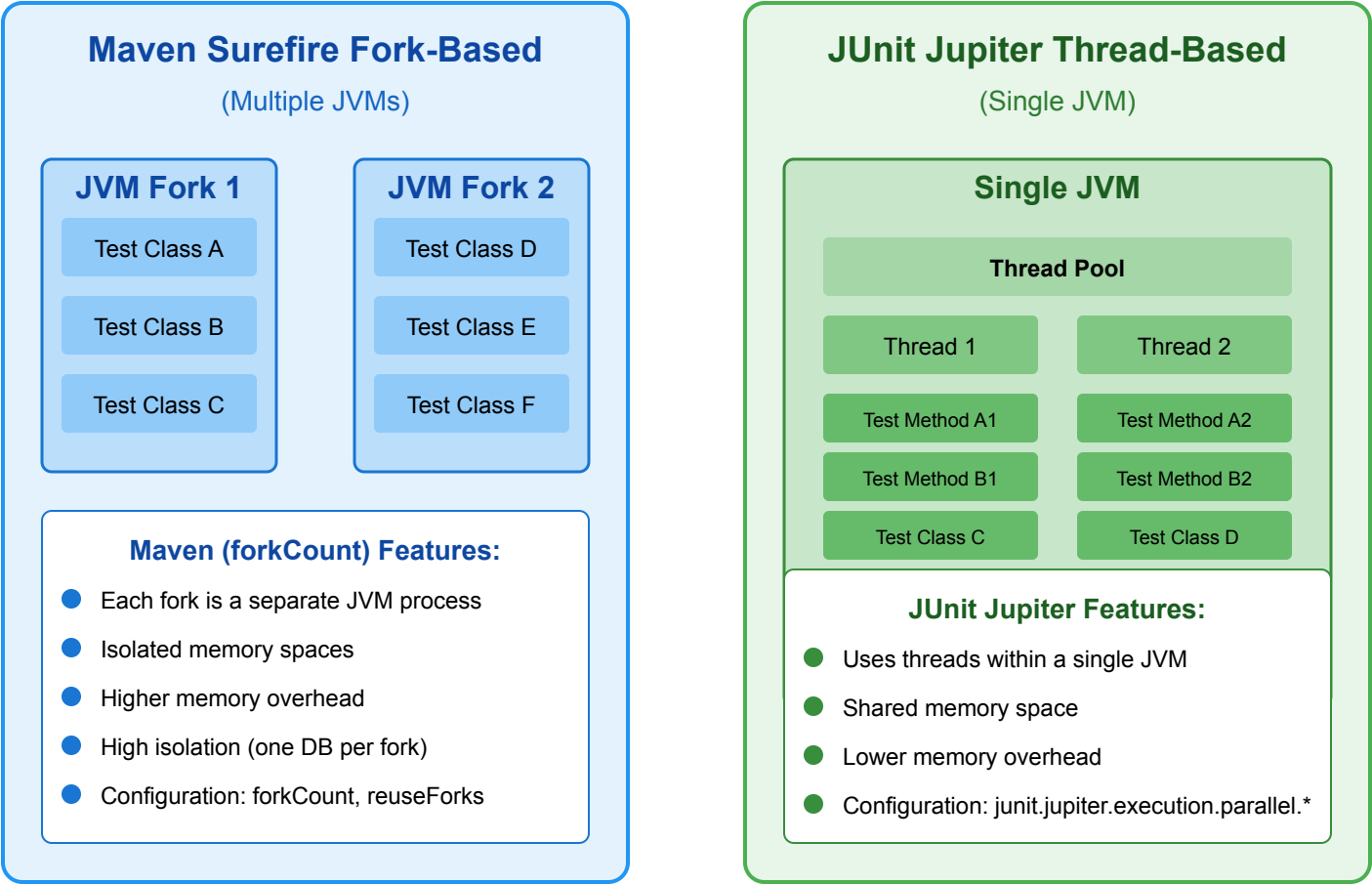
- No shared state
- No dependency between tests and their execution order
- No mutation of global state

Two ways to achieve this:

- Fork a new JVM with Surefire/Failsafe and let it run in parallel -> more resources but isolated execution
- Use JUnit Jupiter's parallelization mode and let it run in the same JVM with multiple threads



Java Test Parallelization Options



Best Practice 2: Get Help from AI

- [Diffblue Cover](#): AI Agent for unit testing complex (Spring Boot) Java code at scale
- My go-to CLI code agent: Claude Code
- TDD with an LLM?
- (Not AI but still useful) OpenRewrite for [automatic code migrations](#) (e.g. JUnit 4 -> JUnit 5)
- Clearly define your requirements in e.g. `claude.md` or Cursor rule files to adopt a common test structure

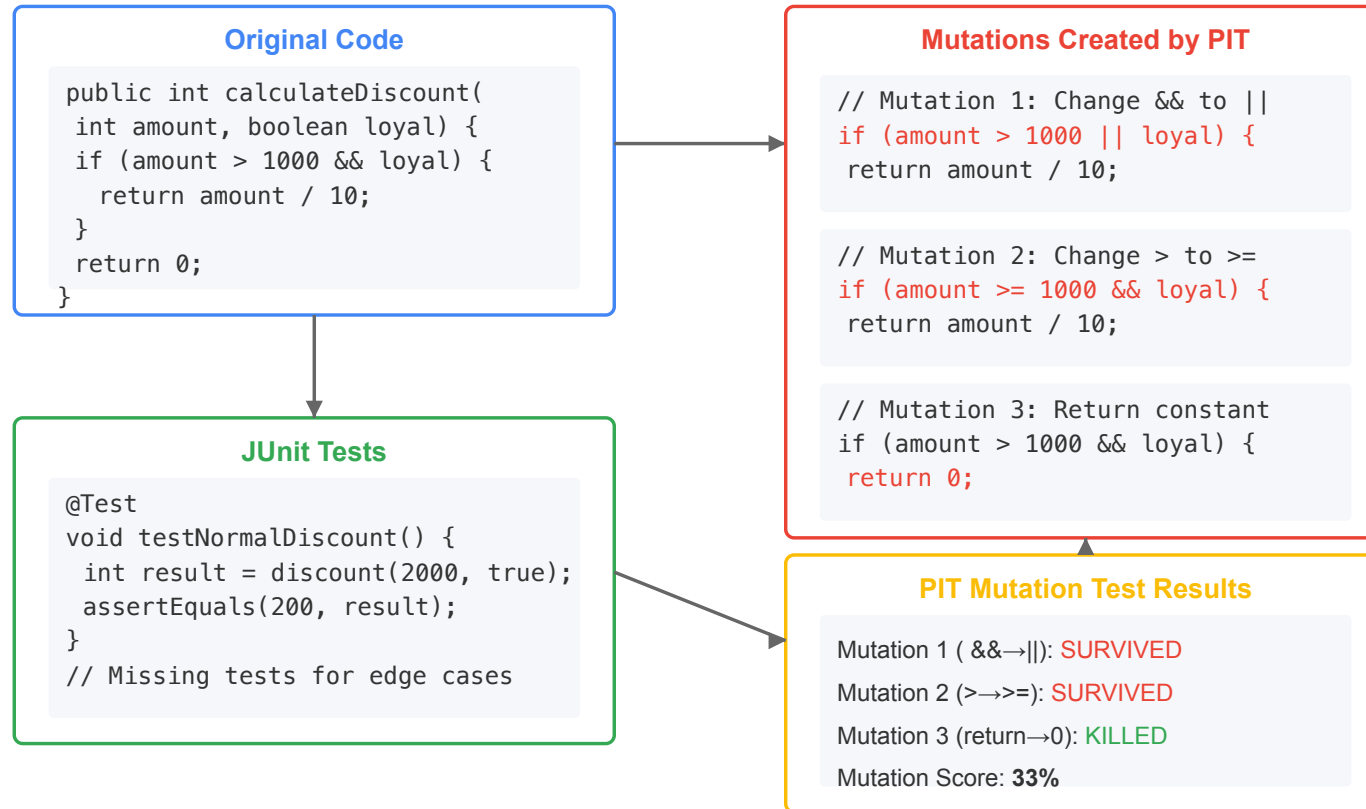


Best Practice 3: Try Mutation Testing

- Having high code coverage might give you a **false sense of security**
- Mutation Testing with [PIT](#)
- Beyond Line Coverage: Traditional tools like JaCoCo show which code runs during tests, but PIT verifies if our tests actually detect when code behaves incorrectly by introducing "**mutations**" to our source code.
- Quality Guarantee: PIT automatically **modifies our code** (changing conditionals, return values, etc.) to ensure our tests fail when they should, **revealing blind spots** in seemingly comprehensive test suites.



PIT Mutation Testing Example



Action Required: Add tests for boundary cases (1000, loyal) and different combinations of inputs



Common Spring Boot Testing Pitfalls to Avoid

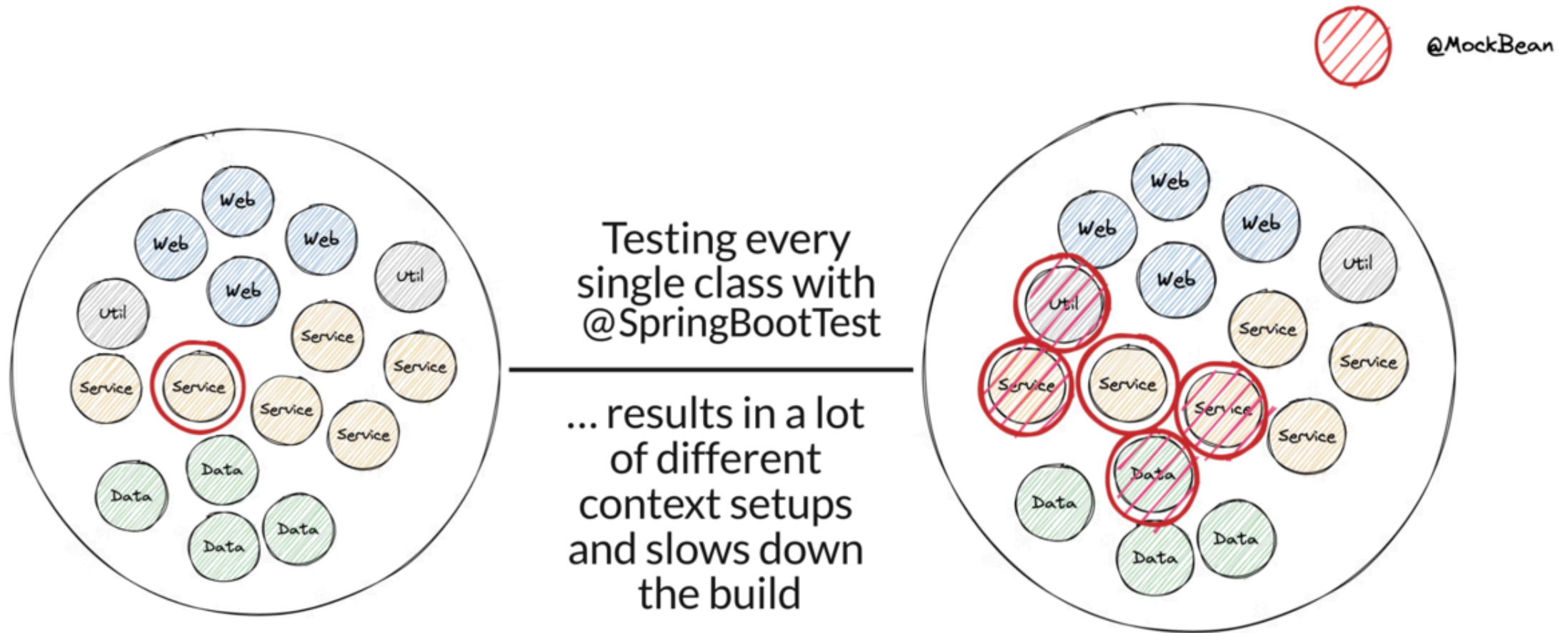


Testing Pitfall 1: `@SpringBootTest` Obsession

- The name could apply it's a one size fits all solution, but it isn't
- It comes with costs: starting the (entire) application context
- Useful for integration tests that verify the whole application but not for testing a single service in isolation
- Start with unit tests, see if sliced tests are applicable and only then use `@SpringBootTest`



@SpringBootTest Obsession Visualized



Testing Pitfall 2: @MockitoBean vs. @MockBean vs. @Mock

- `@MockBean` is a Spring Boot specific annotation that replaces a bean in the application context with a Mockito mock
- `@MockBean` is deprecated in favor of the new `@MockitoBean` annotation
- `@Mock` is a Mockito annotation, only for unit tests
- Golden Mockito Rules:
 - Do not mock types you don't own
 - Don't mock value objects
 - Don't mock everything
 - Show some love with your tests



Testing Pitfall 3: JUnit 4 vs. JUnit 5

- You can mix both versions in the same project but not in the same test class
- Browsing through the internet (aka. StackOverflow/blogs/LLMs) for solutions, you might find test setups that are still for JUnit 4
- Easily import the wrong `@Test` and you end up wasting one hour because the Spring context does not work as expected



JUnit 4	JUnit 5
@Test from org.junit	@Test from org.junit.jupiter.api
@RunWith	@ExtendWith/@RegisterExtension
@ClassRule/@Rule	-
@Before	@BeforeEach
@Ignore	@Disabled
@Category	@Tag



Summary & Outlook

- Spring Boot applications come with batteries-included for testing
- Spring and Spring Boot provides many excellent testing features
- Java provides a mature & rich testing ecosystem
- Consider the context caching feature for fast builds
- Get help from AI
- Still many new testing-related features are part of new releases: pausing a `TestContext`, `@ServiceConnection`, Testcontainers support, Docker Compose support, more AssertJ integrations, etc.



What's Next?

- Online Course: [Testing Spring Boot Applications Masterclass](#) (on-demand, 12 hours, 130+ modules)
- eBook: [30 Testing Tools and Libraries Every Java Developer Must Know](#)
- eBook: [Stratospheric - From Zero to Production with AWS](#)
- Spring Boot [testing workshops](#) (in-house/remote/hybrid)
- [Consulting offerings](#), e.g. the Test Maturity Assessment for projects/teams



Don't Leave Empty-Handed

- Get the complementary **Testing Spring Boot Applications Demystified eBook** for free
- 120+ Pages with hands-on advice to ship code with confidence
- Scan the **QR code on the next slide** to get the free eBook by joining our newsletter



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